

Simplicity Studio UART Rx Code

Goal is to receive the UART message
Assign those values to variables that
can be transmit.

Transmit those values and receive similar
values from similar devices

Store the received values and send them
back through UART.

Also send the RSSI value

Simplicity Studio Program Flow / Pseudocode

Listen For

UART Information - Device ID & match code

Store device ID
and match code

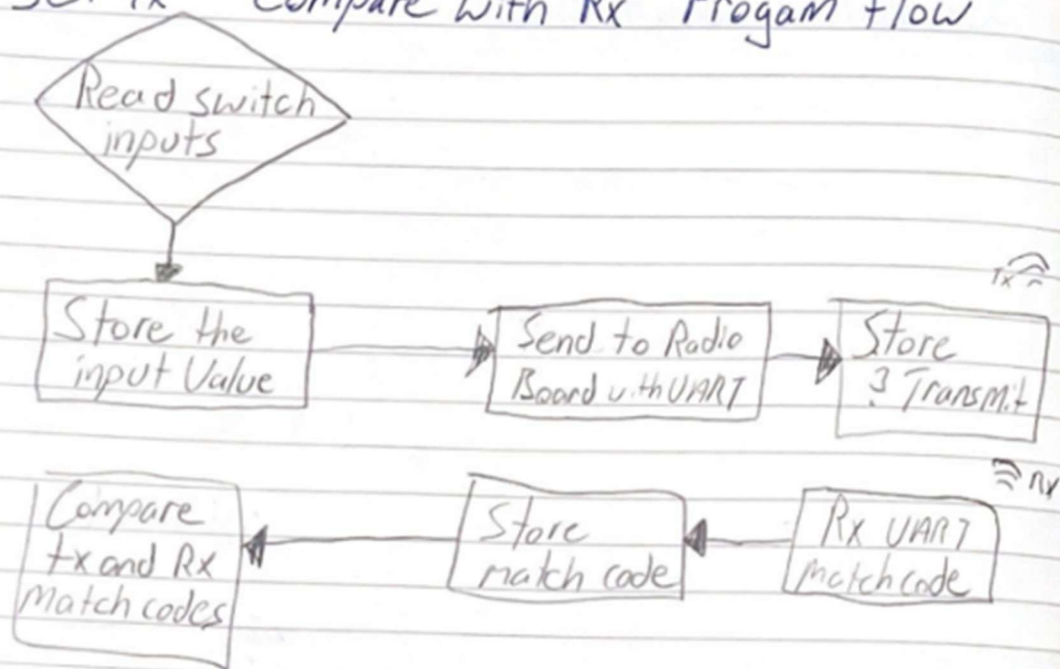
Transmit at 915MHz
include Device ID &
match code in the payload

Listen For Rx Payloads

Store Rx information - Device ID & match code & RSSI

Pass to Pi w. th UART connection

Set Tx Compare with Rx Program Flow



Bitwise &

match code 1	0	6	0000 0110
match code 2	1	4	0001 0100
			& 0000 0100

This is a match of interest
3 out of 8

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Program Flow Address Filtering

Reading Interest 0-7

Assign Address Filter based on Interest
for tx and Rx

Reassign Address Filter depending on Interest
Same addresses for tx and Rx but unique
for each address

Send Interest Value 0-7 through UART
to the radio board

Radio board will send out a specific address mask
and also listen for that same mask

On Rx event communicate to PI through
UART RSSI and Device identification

PI will display ID with location based
on strength

Tx and Rx Code Simplicity Studio 5

Striptesth

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Transmitting a packet

Calling `RAIL_SetTx Fifo(0)`

- Useful for sending almost the same packet with only a few bytes changed

Payload length default 16 bytes
8 bits in a byte

- We need
- 8 bits for match code
 - 8 bits for device ID
 - 8 bits for RSSI

Having trouble changing size of packets sent, can alter payloads

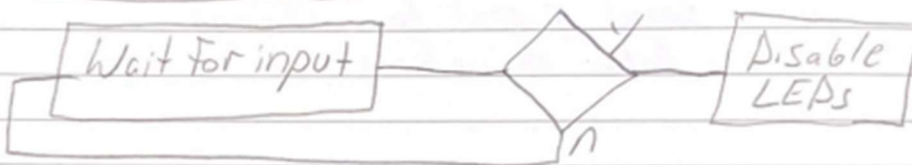
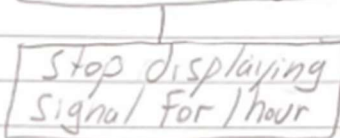
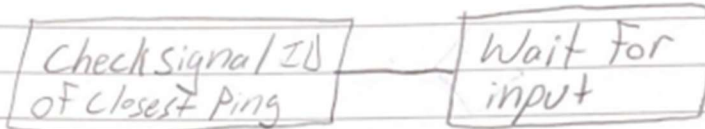
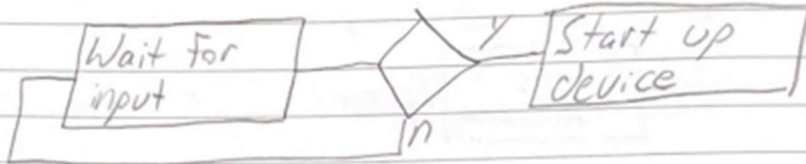
Where to add user code

- `app_init.c` -- initialization
- `app_process.c`

Pseudo Code Buttons

The purpose of the buttons are to use the input to perform a few actions

- Turn device on and off
- Mute the closest ID for 1 hour
- Turn off LEDs



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Updated Program Flow March 13th 2025

